(A405511) DEVOPS LABORATORY

B. Tech (CSE) V Semester

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List of Experiments:

1. Write code for a simple user registration form for an event.
2. Explore Git and GitHub commands.
3. Practice Source code management on GitHub. Experiment with the source code written inexercise 1.
4. Jenkins installation and setup, explore the environment.
5. Demonstrate continuous integration and development using Jenkins.
6. Explore Docker commands for content management.
7. Develop a simple containerized application using Docker.
8. Integrate Kubernetes and Docker
9. Automate the process of running containerized application developed in exercise 7 usingKubernetes.
10. Install and Explore Selenium for automated testing.
11. Write a simple program in JavaScript and perform testing using Selenium.
12. Develop test cases for the above containerized application using selenium.

TEXTBOOKS

1. Joakim Verona. Practical Devops, Second Edition. Ingram short title; 2nd edition (2018). ISBN-10: 1788392574

REFERENCE BOOKS / LEARNING RESOURCES:

1. Deepak Gaikwad, Viral Thakkar. DevOps Tools from Practitioner's Viewpoint. Wiley publications. ISBN: 9788126579952
2. Len Bass, Ingo Weber, Liming Zhu. DevOps: A Software Architect's Perspective. Addison Wesley

COURSE OUTCOMES:

On successful completion of this course, students will be able to:

1. Remember the importance of DevOps tools used in software development life cycle
2. To obtain the complete knowledge of the “version control systems” wth GIt and Git Hub
3. Understand the importance of Jenkins to Build, Deploy and Test Software Applications
4. Analyze & Illustrate the Containerization of OS images and deployment of applications over Docker
5. Create applications and test with various automation tools.

CO-PO MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	1	1	3		1				2			2
CO2	1	1	3		3				3			3
CO3	2	2	3		3				3			2
CO4	1	2	3		3				3			2
CO5	1	2	3		3				2			2

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